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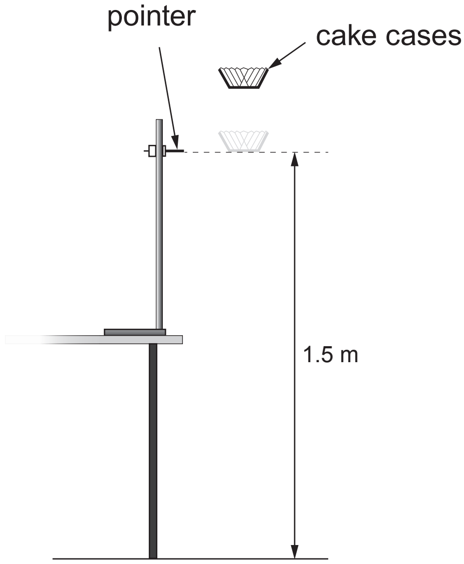
4. A group of students investigate how the terminal speed of falling paper cake cases depends on their mass. They use the apparatus shown in the diagram below.



The cake cases are dropped from at least 20 cm above the pointer. This allows them to reach a terminal speed by the time they reach the pointer. The time taken for the cases to fall the distance of 1.5 m between the pointer and the floor is measured.

The students are given a blank table to record their results. This is shown below.

Number of cake cases	Mass of cake cases (g)	Time taken for paper case cases to fall (s)						Terminal speed (m/s)
		Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Mean	
1								
2								
3								
4								
5								

(a) (i) The same cake cases are used throughout the experiment. State the other controlled **variables** in the experiment. [1]

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(ii) Explain the advantage of 5 trial timings for each number of cases. [2]

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(iii) The students are told the mean mass of a paper case. Explain how the quality of results collected is affected if the cases used each time are weighed rather than calculated by using the mean mass of a case. [2]

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(b) (i) Explain, in terms of forces, how the cake cases reach a terminal speed. [2]

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(ii) Describe how the experiment could be extended to check that the cases are falling with a terminal speed over the 1.5 m shown. [2]

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(c) The cake cases used are supplied in a box containing 250 cases. The combined mass of the box **and** cases is 118 g. The mass of the empty box is 18 g. The terminal speed of 4 paper cases is found to be 2.54 m/s.
The weight of a 1 kg mass on Earth is 10 N.

Use equations from page 2 to calculate:

(i) the time taken for the 4 paper cases to fall the 1.5 m shown in the diagram on page 9. [2]

Time = ... s

(ii) the air resistance acting on 4 paper cases when they fall at terminal speed.
Show your working. [4]

Air resistance = ... N

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